

Sara Klaasen

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Education

PhD	ETH Zurich , Geophysics • ETH medal	Zurich, Switzerland 2020 – 2025
Msc	ETH Zurich , Geophysics	Zurich, Switzerland 2018 – 2020
Bsc	UCR , Liberal Arts and Sciences • Magna cum Laude • Major in Earth sciences, biology, mathematics and physics	Middelburg, Netherlands 2014 – 2017

Experience

Research School of Earth Sciences, Australian National University , Postdoctoral researcher Researching DAS projects in New Zealand. • Imaging Taupo Volcano.	Canberra, Australia 2026 – present 1 year
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland 2025 – 2026 1 year

Awards

ETH Medal The ETH Medal is given to outstanding doctoral theses. ETH Zurich	2026
OSPA Outstanding student presentation award Seismological Society of America Photonics Meeting	2024

Grants

SNF postdoc.mobility grant A grant to fund a 2-year postdoc position. Swiss National Science Foundation	2025
VSBfonds grant A grant to support a Masters programme abroad. VSBfonds	2018

Books

Fiber optic sensing for volcano monitoring and imaging volcanic processes Book chapter in Modern Volcano Monitoring Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, Sara Klaasen, Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk doi.org/10.1007/978-3-031-86841-2_17 (Springer)	2025
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Publications

- Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano** 2026
Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner
doi.org/10.26443/seismica.v5i1.1845 (Seismica)
- Seismic constraints on glacier density** 2025
Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner
doi.org/10.1038/s41598-025-26440-z (Scientific Reports)
- The Seismogenic Thickness of Venus** 2025
Julia Maia, Ana-Catalina PLesá, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia
doi.org/10.1038/s41598-025-26440-z (Journal of Geophysical Research: Planets)
- Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks** 2025
Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner
doi.org/10.1093/gji/ggae459 (Geophysical Journal International)
- An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology** 2024
Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel
dx.doi.org/10.1016/j.eq.s.2024.09.006 (Earthquake Sciences)
- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers** 2024
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina PLesá, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli
doi.org/10.1029/2024EA003670 (Earth and Space Science)
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables** 2024
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti
doi.org/10.1093/gji/ggae364 (Geophysical Journal International)
- Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments** 2024
Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu
doi.org/10.4430/bgo00458 (Bulletin of geophysics and oceanography)
- Challenges in submarine fiber-optic earthquake monitoring** 2024
Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner
doi.org/10.1029/2024JB029556 (Journal of Geophysical Research: Solid Earth)

Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland	2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner doi.org/10.30909/vol.06.02.301311 (Volcanica)	
Fiber-optic observation of volcanic tremor through floating ice sheet resonance	2022
Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir doi.org/10.1785/0320220010 (The Seismic Record)	
Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia	2021
Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner doi.org/10.1029/2021JB022358 (Journal of Geophysical Research: Solid Earth)	
Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing	2021
Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner doi.org/10.1002/essoar.10509693.1 (ESS Open Archive)	
Conference contributions (selected)	
Full-Waveform Inversion for source and structure of DAS data on a subglacial volcano: Grímsvötn, Iceland	2025
invited talk Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner (IASPEI-IGA Scientific Assembly)	
The potential of distributed acoustic sensing for volcano monitoring and imaging	2025
Keynote talk Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Lars Gebraad, Thomas Hudson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jan Dettmer, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece	2025
Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece	2025
Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	
Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland	2025
Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner (IAVCEI General Meeting)	
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece	2025
Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (SSA General Meeting)	

Fibre-Optic Seismology on Remote, Subglacial, Submarine and Actively-Erupting Volcanoes invited talk, outstanding student presentation award Sara Klaasen, Kristin Jonsdottir, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Lars Gebraad, Andreas Fichtner (SSA photonic seismology)	2024
Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU Copernicus Meetings)	2024
Submarine volcano monitoring with fiber-optic cables at Santorini Sara Klaasen, Paraskevi Nomikou, Konstantinos Karantzas, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023
Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023
Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner (IUGG General Assembly)	2023
Monitoring glacially covered volcanoes with Fibre-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (IAVCEI General Meeting)	2023
Volcanic Event Characterization with Fiber-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (AGU Annual Meeting)	2022
Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU General Assembly)	2022
Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland Sara Klaasen, Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner (AGU Annual Meeting)	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (SSA General Meeting)	2021

Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia 2021
Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner
(EGU General Assembly)

Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada 2020
Sara Klaasen, Andreas Fichtner, Patrick Paitz, Jan Dettmer
(AGU Annual Meeting)

Talks (invited)

Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada 2020
Sara Klaasen, Andreas Fichtner, Patrick Paitz, Jan Dettmer
(AGU Annual Meeting)

Languages

Dutch
native

English
C2

German
B2

French
B1